

Homework 7

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Problem 7.1.

Solution 7.1.

1. The matrix is invertible, so the solution set (kernel) is $\mathbf{0}$, the basis is $\emptyset$.
2. By Gaussian Elimination,

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

That is,

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{cases} x - z = 0 \\ y + 2z = 0 \end{cases} \Rightarrow \begin{cases} x = z \\ y = -2z \end{cases}$$

So the kernel is $\left\{ \begin{bmatrix} z \\ -2z \\ z \end{bmatrix}, z \in \mathbb{R} \right\}$, the basis is $\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$.

You can also use the method in the Lecture notes. (See Example 5.7.1.).

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & 2 \\ 0 & 0 & -1 \end{bmatrix}$$

Hence the basis is $\begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}$.

3. Basis of Kernel $\begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ -3 \\ 0 \\ 1 \\ 0 \end{bmatrix}$.

4. Basis of Kernel $\begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 0 \\ 1 \\ 0 \end{bmatrix}$.

5. Basis of Kernel $\begin{bmatrix} 4 \\ -2 \\ 2 \\ -3 \\ 7 \end{bmatrix}$.

6. Basis of Kernel $\begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$.

7. Basis of Kernel is the columns of matrix $\begin{bmatrix} I_n \\ -I_n \end{bmatrix}$.

8. Basis of Kernel is the columns of matrix $\begin{bmatrix} I_n & I_n \\ -I_n & O \\ O & -I_n \end{bmatrix}$.

9. Basis of Kernel is the columns of matrix $\begin{bmatrix} I_n \\ -I_n \end{bmatrix}$.

Problem 7.2.

Solution 7.2.

1. The basis is $(1, \frac{1}{x}, x \sin \frac{1}{x})$. Note that

$$\lim_{x \rightarrow \infty} 1 = 1, \quad \lim_{x \rightarrow \infty} \frac{1}{x} = 0, \quad \lim_{x \rightarrow \infty} x \sin \frac{1}{x} = 1.$$

Hence, the map sends 1 (coordinate: $\mathbf{e}_1$) to 1, sends $\frac{1}{x}$ (coordinate: $\mathbf{e}_2$) to 0, sends $x \sin \frac{1}{x}$ (coordinate: $\mathbf{e}_3$) to 1. Then the matrix is $\begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$.

2. V has only one basis, which is $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. And T sends $\begin{bmatrix} 1 \end{bmatrix}$ to $\begin{bmatrix} -1 \end{bmatrix}$, then $T = \begin{bmatrix} -1 \end{bmatrix}$.

3. Recall that

$$1' = 0, \quad (x+1)' = 1, \quad (x^2)' = 2x.$$

The basis is $(1, x+1, x^2)$. So the map sends $\mathbf{e}_1$ to $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$, sends $\mathbf{e}_2$ to $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, sends $\mathbf{e}_3$ to $\begin{bmatrix} -2 \\ 2 \\ 0 \end{bmatrix}$. Then $T = \begin{bmatrix} 0 & 1 & -2 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$.

4. The basis is $(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix})$. And the map adds the second row to the first row. So $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ (the coordinates are $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$) remain unchanged, and $\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (the coordinates are $\mathbf{e}_4, \mathbf{e}_5, \mathbf{e}_6$) will be mapped to $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$,

whose coordinates are $\mathbf{e}_1 + \mathbf{e}_4, \mathbf{e}_2 + \mathbf{e}_5, \mathbf{e}_3 + \mathbf{e}_6$, respectively. Then the matrix is $\begin{bmatrix} 1 & & & & & \\ & 1 & & & & \\ & & 1 & & & \\ & & & 1 & & \\ & & & & 1 & \\ & & & & & 1 \end{bmatrix}$.

Problem 7.3.

Solution 7.3.

$$1. I_{B \leftarrow C} = \begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 1 & 4 & 12 \\ 0 & 0 & 1 & 6 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \text{ So } I_{C \leftarrow B} = I_{B \leftarrow C}^{-1} = \begin{bmatrix} 1 & -2 & 4 & -8 \\ 0 & 1 & -4 & 12 \\ 0 & 0 & 1 & -6 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

$$2. \text{ Set } \mathcal{D} = (\sin x, \cos x), \text{ then } I_{\mathcal{D} \leftarrow \mathcal{B}} = \begin{bmatrix} 1 & \cos 1 \\ 0 & \sin 1 \end{bmatrix}, I_{\mathcal{D} \leftarrow \mathcal{C}} = \begin{bmatrix} \cos 2 & \cos 3 \\ \sin 2 & \sin 3 \end{bmatrix}.$$

$$\text{Hence } I_{C \leftarrow B} = I_{\mathcal{D} \leftarrow C}^{-1} I_{\mathcal{D} \leftarrow B} = \begin{bmatrix} \cos 2 & \cos 3 \\ \sin 2 & \sin 3 \end{bmatrix}^{-1} \begin{bmatrix} 1 & \cos 1 \\ 0 & \sin 1 \end{bmatrix}.$$

3.

$$I_{C \leftarrow B} \mathbf{e}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{bmatrix}, I_{C \leftarrow B} \mathbf{e}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, I_{C \leftarrow B} \mathbf{e}_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, I_{C \leftarrow B} \mathbf{e}_4 = \begin{bmatrix} 4 \\ 5 \\ 6 \\ 0 \\ 0 \\ 0 \end{bmatrix}, I_{C \leftarrow B} \mathbf{e}_5 = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 0 \\ 4 \\ 5 \end{bmatrix}, I_{C \leftarrow B} \mathbf{e}_6 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 4 \end{bmatrix}. \text{ So}$$

$$I_{C \leftarrow B} = \begin{bmatrix} 1 & 1 & 0 & 4 & 0 & 0 \\ 2 & 0 & 1 & 5 & 1 & 0 \\ 3 & 0 & 0 & 6 & 2 & 1 \\ 4 & 0 & 0 & 0 & 0 & 0 \\ 5 & 1 & 0 & 0 & 4 & 0 \\ 6 & 0 & 1 & 0 & 5 & 4 \end{bmatrix}.$$

4. Note that

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & \\ & 1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 & \\ & -1 \end{bmatrix} \implies I_{C \leftarrow B} \mathbf{e}_1 = \begin{bmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{2} \end{bmatrix}.$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & \\ & 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 & \\ & -1 \end{bmatrix} \implies I_{C \leftarrow B} \mathbf{e}_2 = \begin{bmatrix} \frac{1}{2} \\ 0 \\ -\frac{1}{2} \end{bmatrix}.$$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} - \begin{bmatrix} 1 & \\ & 1 \end{bmatrix} \implies I_{\mathcal{C} \leftarrow \mathcal{B}} e_3 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}.$$

$$\text{Then } I_{\mathcal{C} \leftarrow \mathcal{B}} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -1 \\ 0 & 0 & 1 \\ \frac{1}{2} & -\frac{1}{2} & 0 \end{bmatrix}.$$

5. Set $B = [\mathbf{b}_1 \ \mathbf{b}_2 \ \cdots \ \mathbf{b}_n]$, and $\mathbf{x}_i$ is the coordinates of $\mathbf{b}_i$ under basis $\mathcal{C}$. So $C\mathbf{x}_i = \mathbf{b}_i \implies \mathbf{x}_i = C^{-1}\mathbf{b}_i$. Since the coordinate of $\mathbf{b}_i$ is $\mathbf{e}_i$ under basis $\mathcal{B}$, the i -th column of $I_{\mathcal{C} \leftarrow \mathcal{B}}$ is $\mathbf{x}_i$. Then $I_{\mathcal{C} \leftarrow \mathcal{B}} = [C^{-1}\mathbf{b}_1 \ C^{-1}\mathbf{b}_2 \ \cdots \ C^{-1}\mathbf{b}_n] = C^{-1}B$.

Problem 7.4.

Solution 7.4.

1. The column of A is the linear combination of $\begin{bmatrix} 1 \\ 6 \\ 9 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \\ 8 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, and they are linearly independent, so they are basis of $\text{Ran}(A)$.

Similarly, the basis of $\text{Ran}(A^T)$ is $\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \end{bmatrix}$.

Since $\begin{bmatrix} 1 & & \\ 6 & 1 & \\ 9 & 8 & 1 \end{bmatrix}$ is invertible, then Kernel of A is the same as the Kernel of $\begin{bmatrix} 1 & 2 & 3 & 4 \\ & 1 & 2 & 3 \\ & & 1 & 2 \end{bmatrix}$. The basis is $\begin{bmatrix} 0 \\ -1 \\ 2 \\ -1 \end{bmatrix}$.

Since all the columns of A^T are linearly independent, $A^T \mathbf{x} = \mathbf{0}$ has unique solution, that is, $\mathbf{x} = \mathbf{0}$. The basis of Kernel of A^T is $\emptyset$.

Problem 7.5.

Solution 7.5.

Note that $A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$, so the first column of A is $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$. Also note that $A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, then the second column of A is $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$. Hence $A = \begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix}$.

Problem 7.6.

Solution 7.6.

1. The map sends $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ to $\begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix}$ (coordinate: $\begin{bmatrix} 1 \\ 0 \\ 3 \\ 0 \end{bmatrix}$), sends $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ to $\begin{bmatrix} 0 & 1 \\ 0 & 3 \end{bmatrix}$ (coordinate: $\begin{bmatrix} 0 \\ 1 \\ 0 \\ 3 \end{bmatrix}$), sends $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ to

$\begin{bmatrix} 2 & 0 \\ 4 & 0 \end{bmatrix}$ (coordinate: $\begin{bmatrix} 2 \\ 0 \\ 4 \\ 0 \end{bmatrix}$), sends $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ to $\begin{bmatrix} 0 & 2 \\ 0 & 4 \end{bmatrix}$ (coordinate: $\begin{bmatrix} 0 \\ 2 \\ 0 \\ 4 \end{bmatrix}$)

Then the matrix is $\begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 \\ 3 & 0 & 4 & 0 \\ 0 & 3 & 0 & 4 \end{bmatrix} = \begin{bmatrix} I_2 & 2I_2 \\ 3I_2 & 4I_2 \end{bmatrix}$.

2. The map sends $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ to $\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$ (coordinate: $\begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix}$), sends $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ to $\begin{bmatrix} 3 & 4 \\ 0 & 0 \end{bmatrix}$ (coordinate: $\begin{bmatrix} 3 \\ 4 \\ 0 \\ 0 \end{bmatrix}$), sends $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ to

$\begin{bmatrix} 0 & 0 \\ 1 & 2 \end{bmatrix}$ (coordinate: $\begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \end{bmatrix}$), sends $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ to $\begin{bmatrix} 0 & 0 \\ 3 & 4 \end{bmatrix}$ (coordinate: $\begin{bmatrix} 0 \\ 0 \\ 3 \\ 4 \end{bmatrix}$)

Then the matrix is $\begin{bmatrix} 1 & 3 & 0 & 0 \\ 2 & 4 & 0 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 2 & 4 \end{bmatrix} = \begin{bmatrix} A^T & \\ & A^T \end{bmatrix}$.

3.

$$L_A R_A = \begin{bmatrix} A^T & 2A^T \\ 3A^T & 4A^T \end{bmatrix} = R_A L_A$$

4. multiplying A to X from the right first then multiplying A to XA from the left is equivalent to multiplying A to X from the left first then multiplying A to AX from the right.

5. New Basis $\mathcal{C}$: $C_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $C_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $C_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $C_4 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

Old Basis $\mathcal{B}$: $X_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $X_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $X_3 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, $X_4 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$.

Then $I_{\mathcal{C} \leftarrow \mathcal{B}} = \begin{bmatrix} \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 & -\frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$.

Problem 7.7.

Solution 7.7.

1. True.
2. False. In the left-hand side, the input is coordinates under basis 1, but in the right-hand side, the input is coordinates under basis 2.
3. True. $I_{2 \leftarrow 1} L_{1 \leftarrow 1} I_{2 \leftarrow 1}^{-1} = I_{2 \leftarrow 1} L_{1 \leftarrow 1} I_{1 \leftarrow 2} = L_{2 \leftarrow 2}$.
4. False. The inputs will go through L twice in the right-hand side.
5. True.